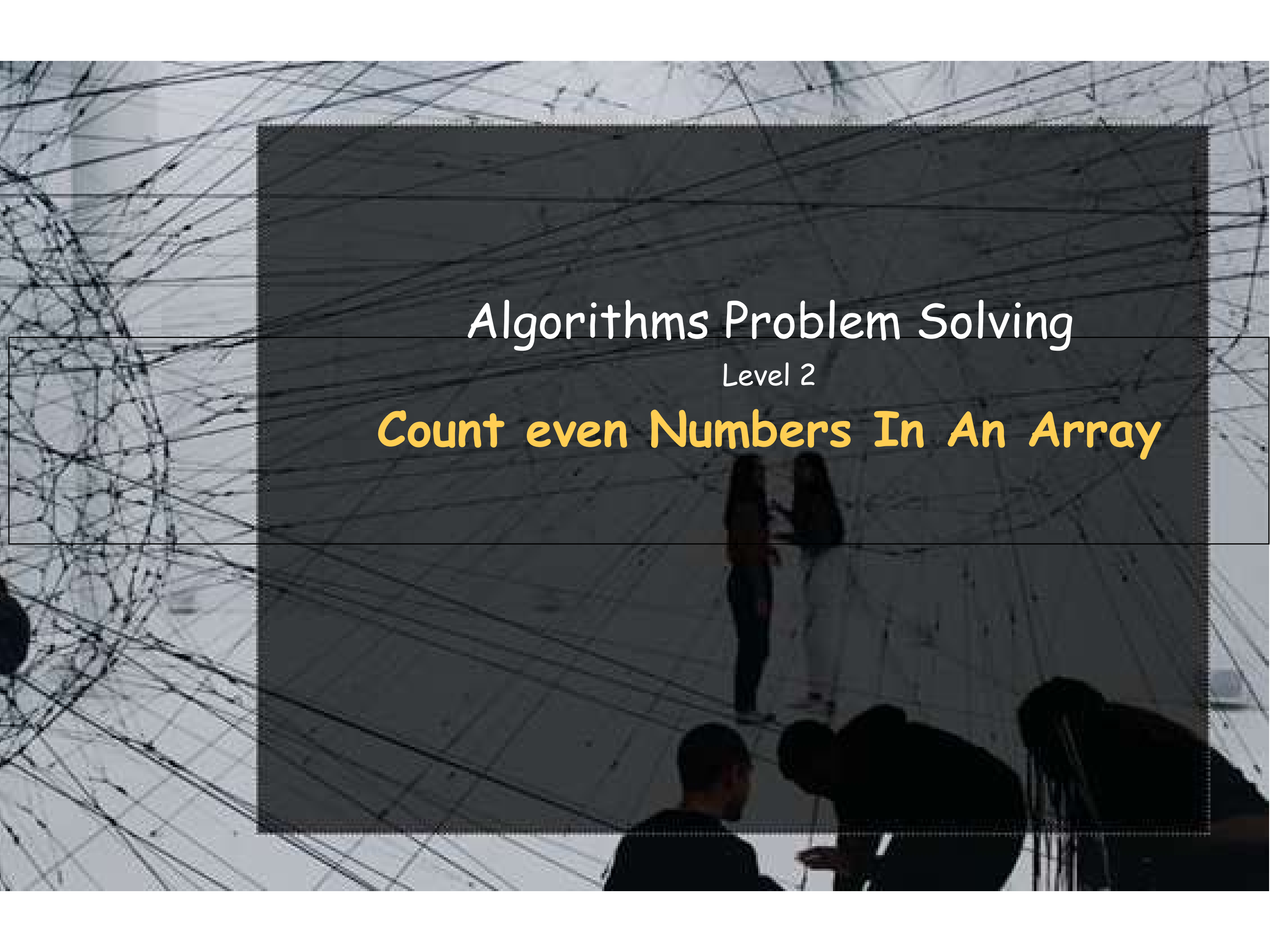


Algorithms Problem Solving Level 2

Count even Numbers In An Array

The background of the slide is a photograph of a modern building's interior. It features a large, curved glass ceiling that allows natural light to filter through, creating a pattern of light and shadow on the floor. In the lower portion of the image, the silhouettes of several people are visible, some standing and others sitting, suggesting a collaborative workspace or a public area. The overall color palette is muted, with greys, blues, and the warm tones of the light coming through the glass.

Algorithms Problem Solving

Level 2

Count even Numbers In An Array

Definition

- ◆ This program fills an array with random numbers from `1` to `100`.
- ◆ It counts how many even numbers exist in the array.
- ◆ Finally, it prints the array elements and the total count of even numbers.

Explanation

- ◆ ``FillArrayWithRandomNumber()`` fills the array with random numbers from ``1`` to ``100``.
- ◆ ``CheckOddOrEven()`` checks whether a number is odd or even using the ``%`` operator.
- ◆ If ``Number % 2 == 0``, the number is even.
- ◆ ``CountEvenNumbers()`` loops through the array and counts every even number.
- ◆ The variable ``count`` starts from ``0`` and increases by ``1`` whenever an even number is found.

Steps in main()

1. Ask the user for the number of elements.
2. Fill the array with random numbers from `1` to `100`.
3. Print all array elements.
4. Loop through the array.
5. Check each number using `CheckOddOrEven()`.
6. If the number is even, increase `count`.
7. Return the total count.
8. Print the number of even elements.

💡 Example

◆ `Array Elements` :

78 97 32 14 14 42 32 18 55 44

◆ Even numbers are:

78 32 14 14 42 32 18 44

◆ The even numbers are counted one by one:

78 → Even → count = 1

97 → Odd

32 → Even → count = 2

14 → Even → count = 3

14 → Even → count = 4

42 → Even → count = 5

32 → Even → count = 6

18 → Even → count = 7

55 → Odd

44 → Even → count = 8

➡ Result

Array Elements: 78 97 32 14 14 42 32 18 55 44

Even Numbers count is : 8

Note

◆ Important statement:

```
if (CheckOddOrEven(arr[i]) == enOddOrEven::Even)
    count++;
```

◆ `CheckOddOrEven()` determines whether the number is `Odd` or `Even`.

◆ `CountEvenNumbers()` is responsible for counting the even numbers.

◆ `count` stores how many even numbers were found.

◆ `ReadArrayElements()` and `AddElementToArray()` are defined but not used in this program.

⊘ Warning

- ◆ `arr[100]` can store a maximum of `100` elements, so `arrLength` should not be greater than `100`.
- ◆ `FillArrayWithRandomNumber()` does not validate the entered length.
- ◆ `cmath` and `corecrt_math_defines.h` are not needed in this program.
- ◆ `rand()` and `time()` should ideally have their required headers included explicitly.
- ◆ `rand()` generates pseudo-random numbers, not security-grade random numbers.



Comparison

▣ `CountEvenNumbers()`

- ✓ Counts only even numbers.
- ✓ Uses `CheckOddOrEven()` to identify even values.
- ✓ Returns the total number of even elements.

▤ `CountOddNumbers()`

- ✓ Counts only odd numbers.
- ✓ Uses `CheckOddOrEven()` to identify odd values.
- ✓ Returns the total number of odd elements.

✦ Summary

- ✓ Fill the array with random numbers from `1` to `100`.
- ✓ Check every element to determine if it is even.
- ✓ Increase `count` for every even number.
- ✓ Return the total count.
- ✓ Print the number of even elements.

Thank you 😊